STA_445_Assignment 7

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Problem 1:

The `infmort` data set from the package `faraway` gives the infant mortality rate for a variety of countries. The information is relatively out of date, but will be fun to graph. Visualize the data using by creating scatter plots of mortality vs income while faceting using `region` and setting color by `oil` export status. Utilize a $\log_{10}$ transformation for both `mortality` and `income` axes. This can be done either by doing the transformation inside the `aes()` command or by utilizing the `scale_x_log10()` or `scale_y_log10()` layers. The critical difference is if the scales are on the original vs log transformed scale. Experiment with both and see which you prefer.

a. The `rownames()` of the table gives the country names and you should create a new column that contains the country names. *`rownames`

```
(countrynames <- infmort %>% mutate(country = rownames(infmort) %>% str_trim()))
```

```
##                       region income mortality            oil
## Australia               Asia   3426      26.7 no oil exports
## Austria               Europe   3350      23.7 no oil exports
## Belgium               Europe   3346      17.0 no oil exports
## Canada              Americas   4751      16.8 no oil exports
## Denmark               Europe   5029      13.5 no oil exports
## Finland               Europe   3312      10.1 no oil exports
## France                Europe   3403      12.9 no oil exports
## West_Germany          Europe   5040      20.4 no oil exports
## Ireland               Europe   2009      17.8 no oil exports
## Italy                 Europe   2298      25.7 no oil exports
## Japan                 Europe   3292      11.7 no oil exports
## Netherlands           Europe   4103      11.6 no oil exports
## New_Zealand             Asia   3723      16.2 no oil exports
## Norway                Europe   4102      11.3 no oil exports
## Portugal              Europe    956      44.8 no oil exports
## South_Africa          Africa   1000      71.5 no oil exports
## Sweden                Europe   5596       9.6 no oil exports
## Switzerland           Europe   2963      12.8 no oil exports
## Britain               Europe   2503      17.5 no oil exports
## United_States       Americas   5523      17.6 no oil exports
## Algeria               Africa    400      86.3    oil exports
## Ecuador             Americas    250      78.5    oil exports
## Indonesia               Asia    110     125.0    oil exports
## Iran                    Asia   1280        NA    oil exports
```

```
## Iraq                    Asia    560      28.1    oil exports
## Libya                  Africa   3010     300.0    oil exports
## Nigeria                Africa    220      58.0    oil exports
## Saudi_Arabia            Asia   1530     650.0    oil exports
## Venezuela           Americas   1240      51.7    oil exports
## Argentina           Americas   1191      59.6 no oil exports
## Brazil              Americas    425     170.0 no oil exports
## Chile               Americas    590      78.0 no oil exports
## Colombia            Americas    426      62.8 no oil exports
## Costa_Rica          Americas    725      54.4 no oil exports
## Dominican_Republic  Americas    406      48.8 no oil exports
## Greece                 Europe   1760      27.8 no oil exports
## Guatemala           Americas    302      79.1 no oil exports
## Israel                  Asia   2526      22.1 no oil exports
## Jamaica             Americas    727      26.2 no oil exports
## Lebanon                 Asia    631      13.6 no oil exports
## Malaysia                Asia    295      32.0 no oil exports
## Mexico              Americas    684      60.9 no oil exports
## Nicaragua           Americas    507      46.0 no oil exports
## Panama              Americas    754      34.1 no oil exports
## Peru                Americas    335      65.1 no oil exports
## Singapore               Asia   1268      20.4 no oil exports
## Spain                  Europe   1256      15.1 no oil exports
## Taiwan                  Asia    261      19.1 no oil exports
## Trinidad_and_Tobago Americas    732      26.2 no oil exports
## Tunisia                Africa    434      76.3 no oil exports
## Uruguay             Americas    799      40.4 no oil exports
## Yugoslavia             Europe    406      43.3 no oil exports
## Zambia                 Africa    310     259.0 no oil exports
## Bolivia             Americas    200      60.4 no oil exports
## Cameroon               Africa    100     137.0 no oil exports
## Congo                  Africa    281     180.0 no oil exports
## Egypt                  Africa    210     114.0 no oil exports
## El_Salvador         Americas    319      58.2 no oil exports
## Ghana                  Africa    217      63.7 no oil exports
## Honduras            Americas    284      39.3 no oil exports
## Ivory_Coast            Africa    387     138.0 no oil exports
## Jordan                  Asia    334      21.3 no oil exports
## South_Korea             Asia    344      58.0 no oil exports
## Liberia                Africa    197     159.2 no oil exports
## Moroco                 Africa    279     149.0 no oil exports
## Papua_New_Guinea        Asia    477      10.2 no oil exports
## Paraguay            Americas    347      38.6 no oil exports
## Philippines             Asia    230      67.9 no oil exports
## Syria                   Asia    334      21.7 no oil exports
## Thailand                Asia    210      27.0 no oil exports
## Turkey                  Asia    435     153.0 no oil exports
## South_Vietnam           Asia    130     100.0 no oil exports
## Afganistan              Asia     75     400.0 no oil exports
## Bangladesh              Asia    100     124.3 no oil exports
## Burma                   Asia     73     200.0 no oil exports
## Burundi                Africa     68     150.0 no oil exports
## Cambodia                Asia    123     100.0 no oil exports
## Central_African_Rep    Africa    122     190.0 no oil exports
```

```
## Chad                  Africa     70     160.0 no oil exports
## Dahomey               Africa     81     109.6 no oil exports
## Ethiopia              Africa     79      84.2 no oil exports
## Guinea                Africa     79     216.0 no oil exports
## Haiti                Americas    100        NA no oil exports
## India                    Asia     93      60.6 no oil exports
## Kenya                 Africa    169      55.0 no oil exports
## Laos                     Asia     71        NA no oil exports
## Madagascar            Africa    120     102.0 no oil exports
## Malawi                Africa    130     148.3 no oil exports
## Mali                  Africa     50     120.0 no oil exports
## Mauritania            Africa    174     187.0 no oil exports
## Nepal                    Asia     90        NA no oil exports
## Niger                 Africa     70     200.0 no oil exports
## Pakistan                 Asia    102     124.3 no oil exports
## Rwanda                Africa     61     132.9 no oil exports
## Sierra_Leone          Africa    148     170.0 no oil exports
## Somalia               Africa     85     158.0 no oil exports
## Sri_Lanka                Asia    162      45.1 no oil exports
## Sudan                 Africa    125     129.4 no oil exports
## Tanzania              Africa    120     162.5 no oil exports
## Togo                  Africa    160     127.0 no oil exports
## Uganda                Africa    134     160.0 no oil exports
## Upper_Volta           Africa     82     180.0 no oil exports
## Southern_Yemen           Asia     96      80.0 no oil exports
## Yemen                    Asia     77      50.0 no oil exports
## Zaire                 Africa    118     104.0 no oil exports
##                                 country
## Australia                     Australia
## Austria                         Austria
## Belgium                         Belgium
## Canada                           Canada
## Denmark                         Denmark
## Finland                         Finland
## France                           France
## West_Germany               West_Germany
## Ireland                         Ireland
## Italy                             Italy
## Japan                             Japan
## Netherlands                 Netherlands
## New_Zealand                 New_Zealand
## Norway                           Norway
## Portugal                       Portugal
## South_Africa               South_Africa
## Sweden                           Sweden
## Switzerland                 Switzerland
## Britain                         Britain
## United_States             United_States
## Algeria                         Algeria
## Ecuador                         Ecuador
## Indonesia                     Indonesia
## Iran                               Iran
## Iraq                               Iraq
## Libya                             Libya
```

```
## Nigeria                        Nigeria
## Saudi_Arabia               Saudi_Arabia
## Venezuela                     Venezuela
## Argentina                     Argentina
## Brazil                           Brazil
## Chile                             Chile
## Colombia                       Colombia
## Costa_Rica                   Costa_Rica
## Dominican_Republic   Dominican_Republic
## Greece                           Greece
## Guatemala                     Guatemala
## Israel                           Israel
## Jamaica                         Jamaica
## Lebanon                         Lebanon
## Malaysia                       Malaysia
## Mexico                           Mexico
## Nicaragua                     Nicaragua
## Panama                           Panama
## Peru                               Peru
## Singapore                     Singapore
## Spain                             Spain
## Taiwan                           Taiwan
## Trinidad_and_Tobago Trinidad_and_Tobago
## Tunisia                         Tunisia
## Uruguay                         Uruguay
## Yugoslavia                   Yugoslavia
## Zambia                           Zambia
## Bolivia                         Bolivia
## Cameroon                       Cameroon
## Congo                             Congo
## Egypt                             Egypt
## El_Salvador                 El_Salvador
## Ghana                             Ghana
## Honduras                       Honduras
## Ivory_Coast                 Ivory_Coast
## Jordan                           Jordan
## South_Korea                 South_Korea
## Liberia                         Liberia
## Moroco                           Moroco
## Papua_New_Guinea       Papua_New_Guinea
## Paraguay                       Paraguay
## Philippines                 Philippines
## Syria                             Syria
## Thailand                       Thailand
## Turkey                           Turkey
## South_Vietnam             South_Vietnam
## Afganistan                   Afganistan
## Bangladesh                   Bangladesh
## Burma                             Burma
## Burundi                         Burundi
## Cambodia                       Cambodia
## Central_African_Rep Central_African_Rep
## Chad                               Chad
## Dahomey                         Dahomey
```

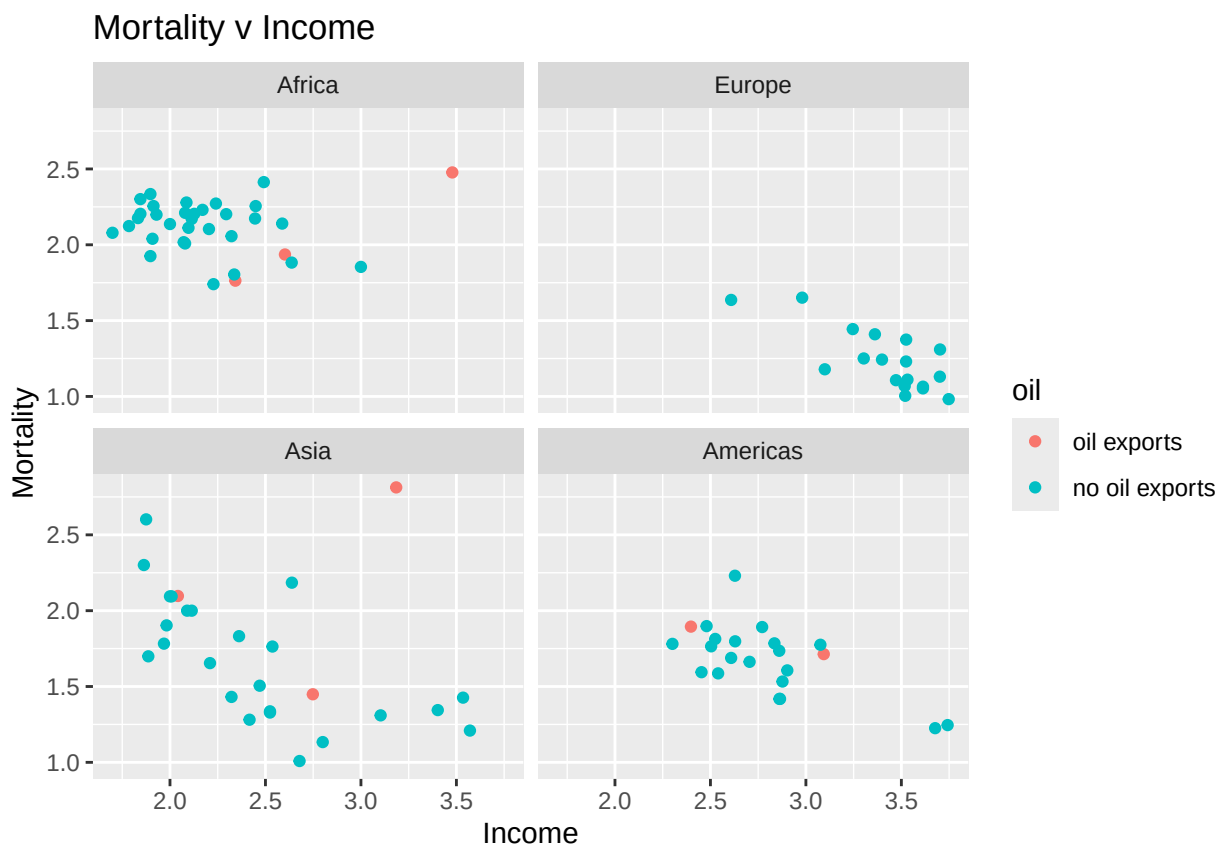
```
## Ethiopia                        Ethiopia
## Guinea                            Guinea
## Haiti                              Haiti
## India                              India
## Kenya                              Kenya
## Laos                                Laos
## Madagascar                   Madagascar
## Malawi                           Malawi
## Mali                                Mali
## Mauritania                   Mauritania
## Nepal                              Nepal
## Niger                              Niger
## Pakistan                       Pakistan
## Rwanda                           Rwanda
## Sierra_Leone                Sierra_Leone
## Somalia                          Somalia
## Sri_Lanka                     Sri_Lanka
## Sudan                              Sudan
## Tanzania                       Tanzania
## Togo                                Togo
## Uganda                           Uganda
## Upper_Volta                  Upper_Volta
## Southern_Yemen            Southern_Yemen
## Yemen                              Yemen
## Zaire                              Zaire
```

b. Create scatter plots with the `log10()` transformation inside the `aes()`command.

```
infplot1 <- ggplot(countrynames, aes(x = log10(income), y = log10(mortality), color = oil)) +
  geom_point() + facet_wrap(~region) + labs(x = "Income", y = "Mortality",
  title = "Mortality v Income")

infplot1
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

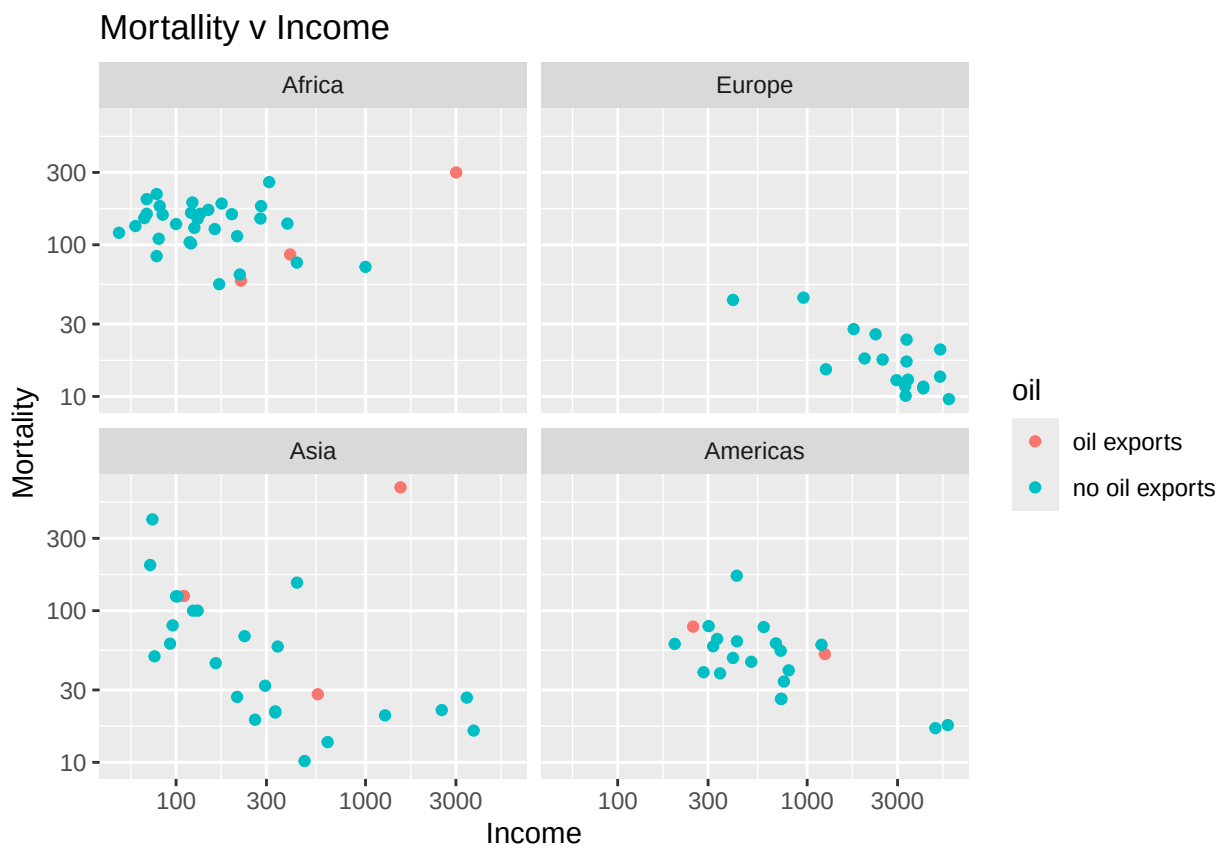


c. Create the scatter plots using the `scale_x_log10()` and `scale_y_log10()`.Set the major and minor breaks to be useful and aesthetically pleasing.Comment on which version you find easier to read.

I think this scale_x_log10 version is easier to read

```
infplot2 <- ggplot(countrynames, aes(x = income, y = mortality, color = oil)) + geom_point() +
  facet_wrap(~region) + scale_x_log10() + scale_y_log10() +
  labs(x = "Income", y = "Mortality", title = "Mortallity v Income")
infplot2
```

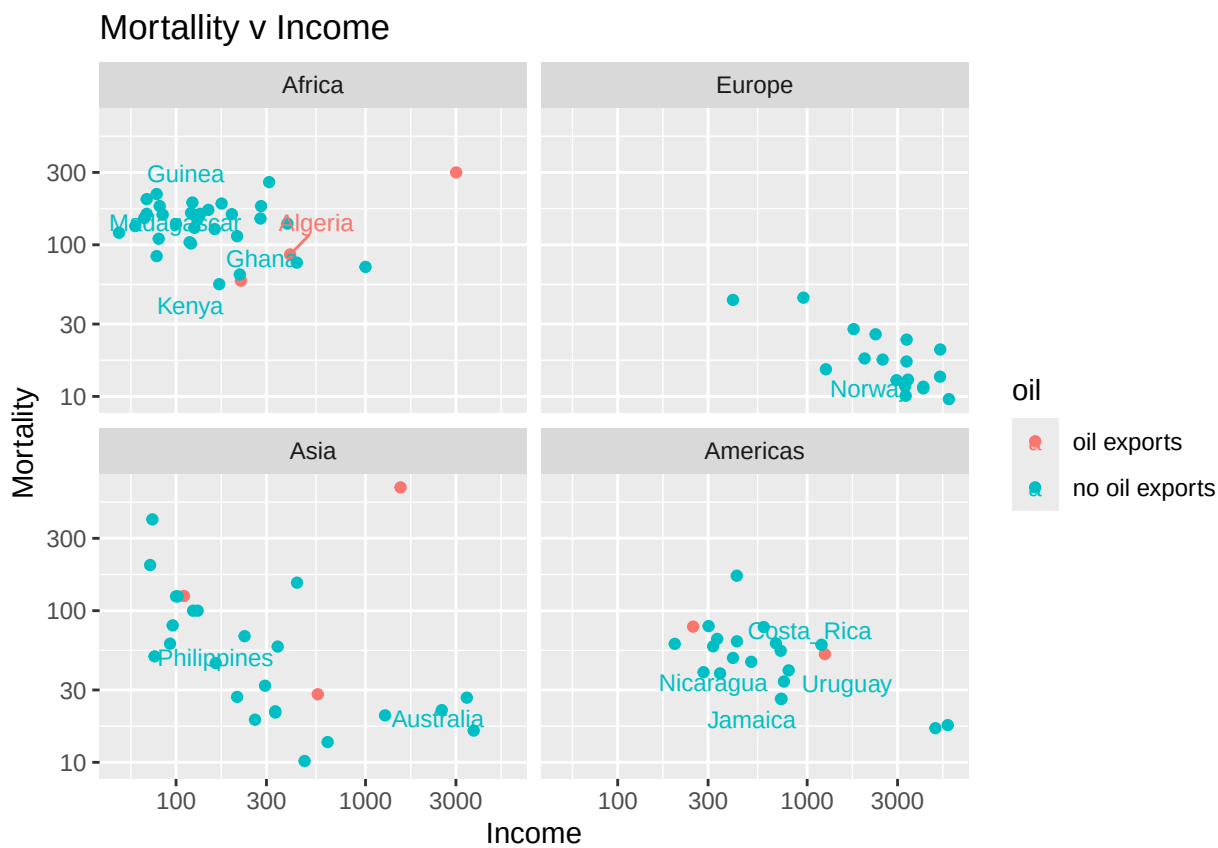
```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



d. The package `ggrepel` contains functions `geom_text_repel()` and `geom_label_repel()` that mimic the basic `geom_text()` and `geom_label()`functions in `ggplot2`, but work to make sure the labels don't overlap. Select 10-15 countries to label and do so using the `geom_text_repel()` function.

```
rand_countries <- countrynames %>% slice_sample(n = 12)
infplot3 <- infplot2 + geom_text_repel(data = rand_countries, aes(label = country), size = 3)
infplot3
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Problem 2

Using the `datasets::trees` data, complete the following:

a. Create a regression model for $y =$ `Volume` as a function of $x =$ `Height`.

```
data(trees)
(regmodel <- lm(Volume ~ Height, data = trees))
```

```
##
## Call:
## lm(formula = Volume ~ Height, data = trees)
##
## Coefficients:
## (Intercept)       Height
##     -87.124        1.543
```

b. Using the str(your model's name) command, to get a list of all the information stored in the linear model object. Use $ to extract the slope and intercept of the regression line (the coefficients).

```
str(regmodel)
```

```
## List of 12
```

```
##  $ coefficients : Named num [1:2] -87.12 1.54
##   ..- attr(*, "names")= chr [1:2] "(Intercept)" "Height"
##  $ residuals    : Named num [1:31] -10.6109 -2.8941 0.0926 -7.5976 -19.0877 ...
##   ..- attr(*, "names")= chr [1:31] "1" "2" "3" "4" ...
##  $ effects      : Named num [1:31] -167.98 53.86 2.05 -5.89 -17.64 ...
##   ..- attr(*, "names")= chr [1:31] "(Intercept)" "Height" "" "" ...
##  $ rank         : int 2
##  $ fitted.values: Named num [1:31] 20.9 13.2 10.1 24 37.9 ...
##   ..- attr(*, "names")= chr [1:31] "1" "2" "3" "4" ...
##  $ assign       : int [1:2] 0 1
##  $ qr           :List of 5
##   ..$ qr   : num [1:31, 1:2] -5.57 0.18 0.18 0.18 0.18 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   .. .. ..$ : chr [1:31] "1" "2" "3" "4" ...
##   .. .. ..$ : chr [1:2] "(Intercept)" "Height"
##   .. ..- attr(*, "assign")= int [1:2] 0 1
##   ..$ qraux: num [1:2] 1.18 1.29
##   ..$ pivot: int [1:2] 1 2
##   ..$ tol  : num 1e-07
##   ..$ rank : int 2
##   ..- attr(*, "class")= chr "qr"
##  $ df.residual  : int 29
##  $ xlevels      : Named list()
##  $ call         : language lm(formula = Volume ~ Height, data = trees)
##  $ terms        :Classes 'terms', 'formula'  language Volume ~ Height
##   .. ..- attr(*, "variables")= language list(Volume, Height)
##   .. ..- attr(*, "factors")= int [1:2, 1] 0 1
##   .. .. ..- attr(*, "dimnames")=List of 2
##   .. .. .. ..$ : chr [1:2] "Volume" "Height"
##   .. .. .. ..$ : chr "Height"
##   .. ..- attr(*, "term.labels")= chr "Height"
##   .. ..- attr(*, "order")= int 1
##   .. ..- attr(*, "intercept")= int 1
##   .. ..- attr(*, "response")= int 1
##   .. ..- attr(*, ".Environment")=<environment: R_GlobalEnv>
##   .. ..- attr(*, "predvars")= language list(Volume, Height)
##   .. ..- attr(*, "dataClasses")= Named chr [1:2] "numeric" "numeric"
##   .. .. ..- attr(*, "names")= chr [1:2] "Volume" "Height"
##  $ model        :'data.frame':   31 obs. of  2 variables:
##   ..$ Volume: num [1:31] 10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 ...
##   ..$ Height: num [1:31] 70 65 63 72 81 83 66 75 80 75 ...
##   ..- attr(*, "terms")=Classes 'terms', 'formula'  language Volume ~ Height
##   .. .. ..- attr(*, "variables")= language list(Volume, Height)
##   .. .. ..- attr(*, "factors")= int [1:2, 1] 0 1
##   .. .. .. ..- attr(*, "dimnames")=List of 2
##   .. .. .. .. ..$ : chr [1:2] "Volume" "Height"
##   .. .. .. .. ..$ : chr "Height"
##   .. .. ..- attr(*, "term.labels")= chr "Height"
##   .. .. ..- attr(*, "order")= int 1
##   .. .. ..- attr(*, "intercept")= int 1
##   .. .. ..- attr(*, "response")= int 1
##   .. .. ..- attr(*, ".Environment")=<environment: R_GlobalEnv>
##   .. .. ..- attr(*, "predvars")= language list(Volume, Height)
##   .. .. ..- attr(*, "dataClasses")= Named chr [1:2] "numeric" "numeric"
```

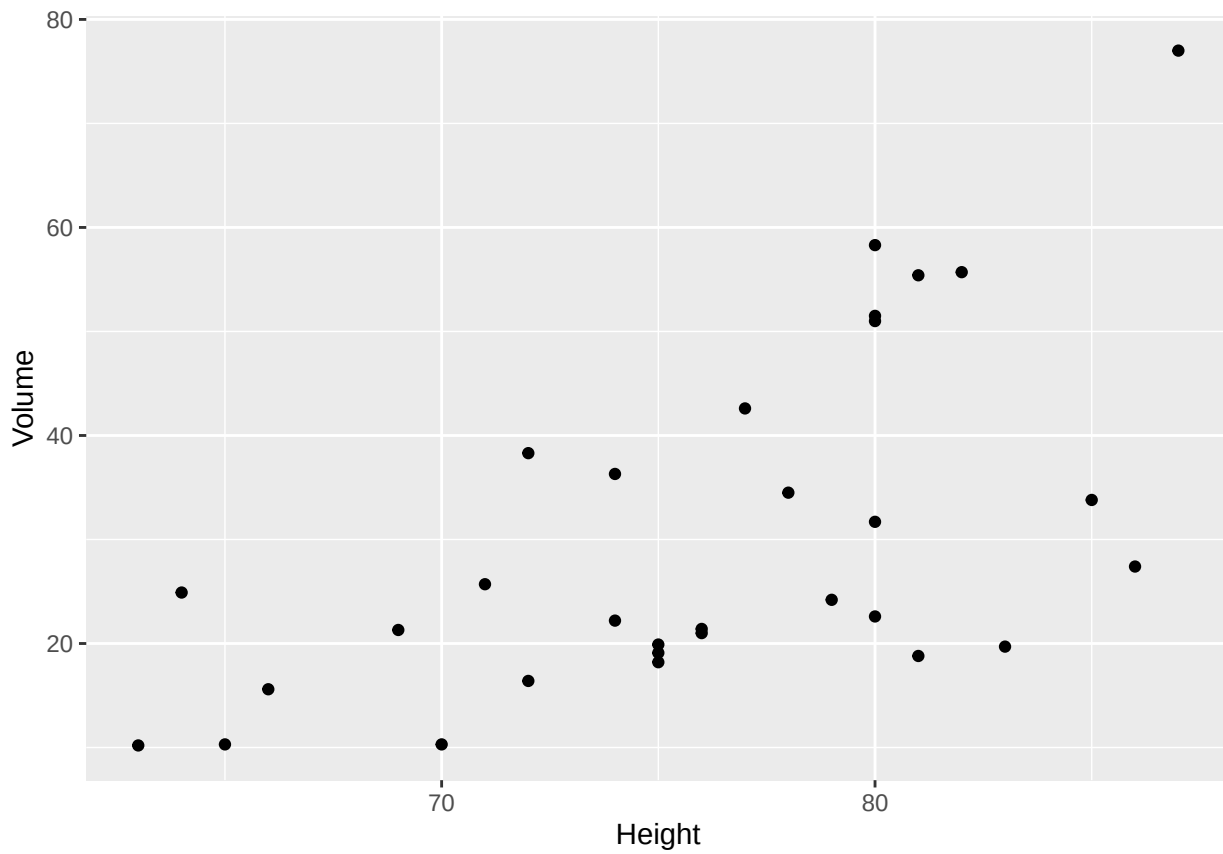
```
##   .. .. .. ..- attr(*, "names")= chr [1:2] "Volume" "Height"
##  - attr(*, "class")= chr "lm"
```

```
regmodel$coefficients
```

```
## (Intercept)      Height
##   -87.12361     1.54335
```

c. Using `ggplot2`, create a scatter plot of Volume vs Height.

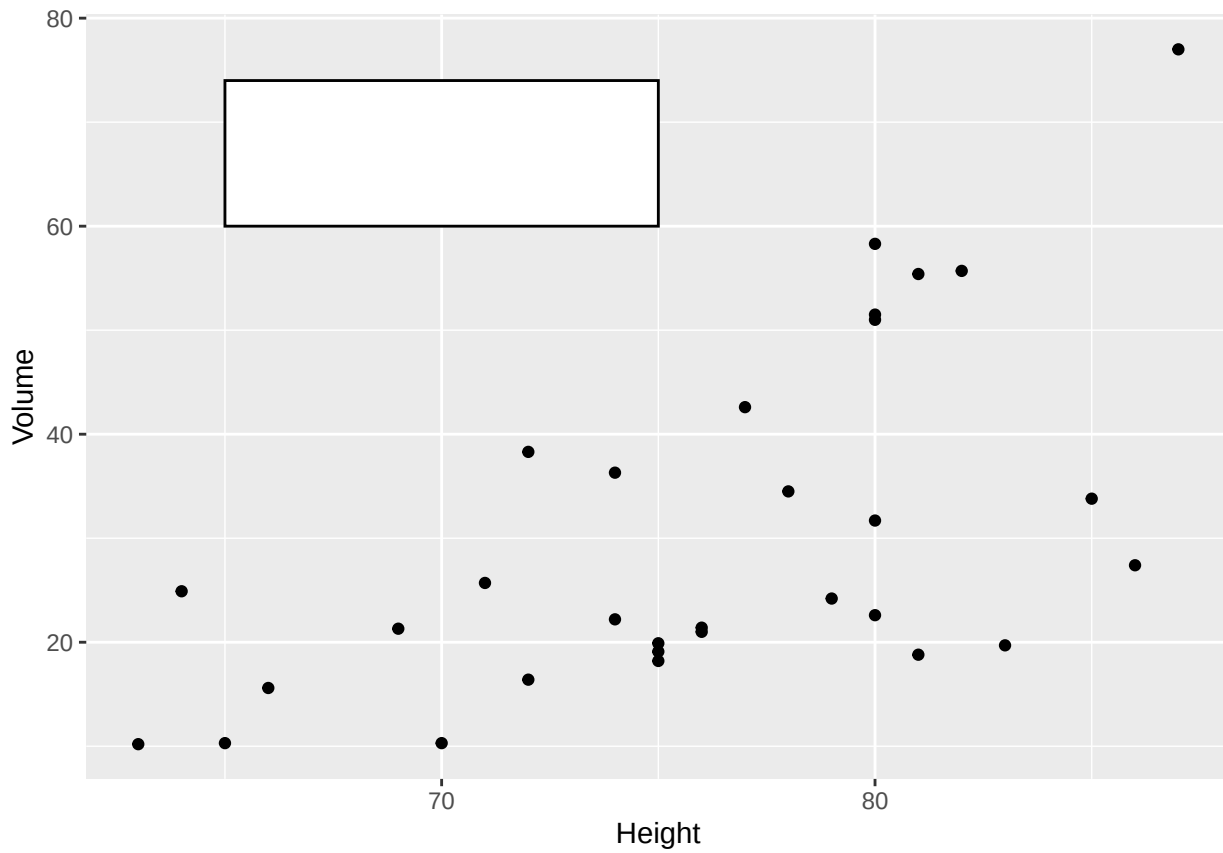
```
PlotA1 <- ggplot(trees, aes(x = Height, y = Volume)) + geom_point()
PlotA1
```



d. Create a nice white filled rectangle to add text information to using by adding the following annotation layer.

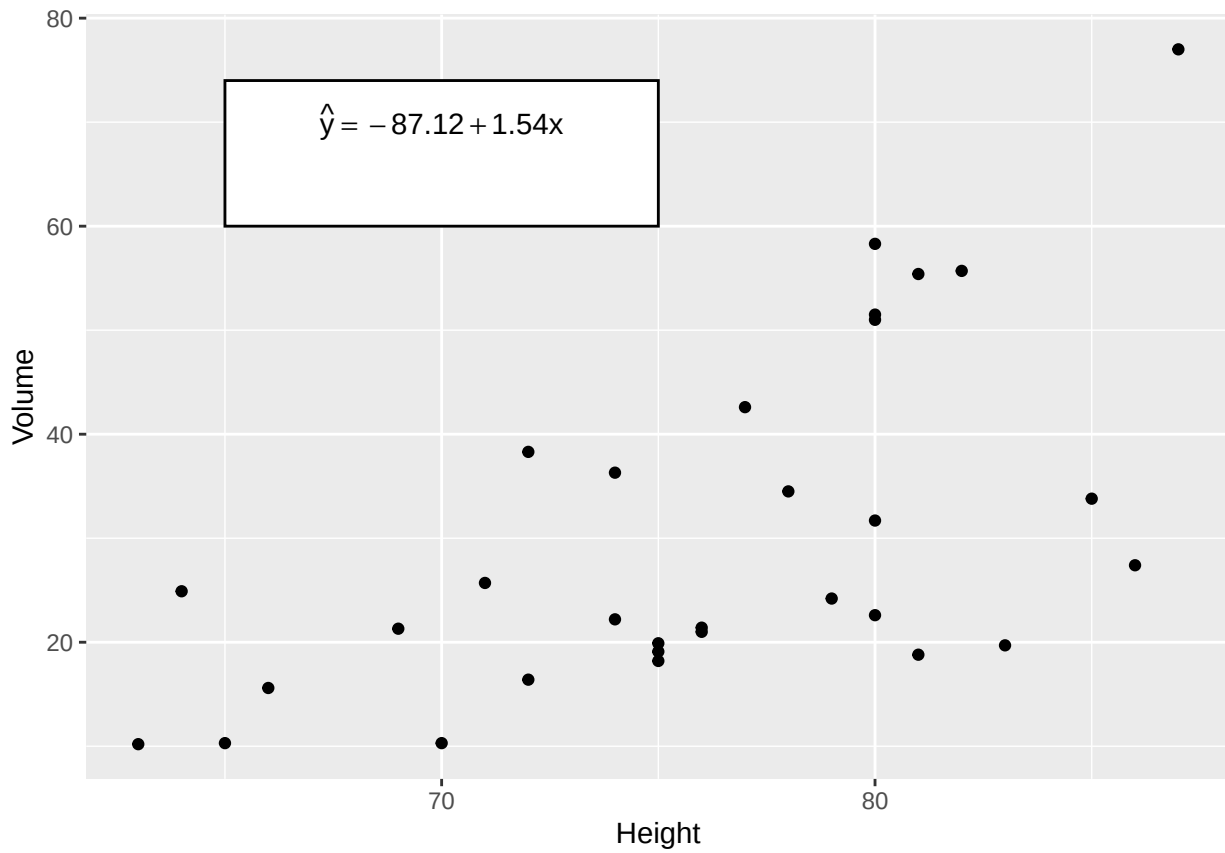
annotate('rect', xmin=65, xmax=75, ymin=60, ymax=74, fill='white', color='black')

```
(plotA2 <- ggplot(trees, aes(x = Height, y = Volume)) + geom_point() +
  annotate('rect', xmin=65, xmax=75, ymin=60, ymax=74, fill='white', color='black'))
```



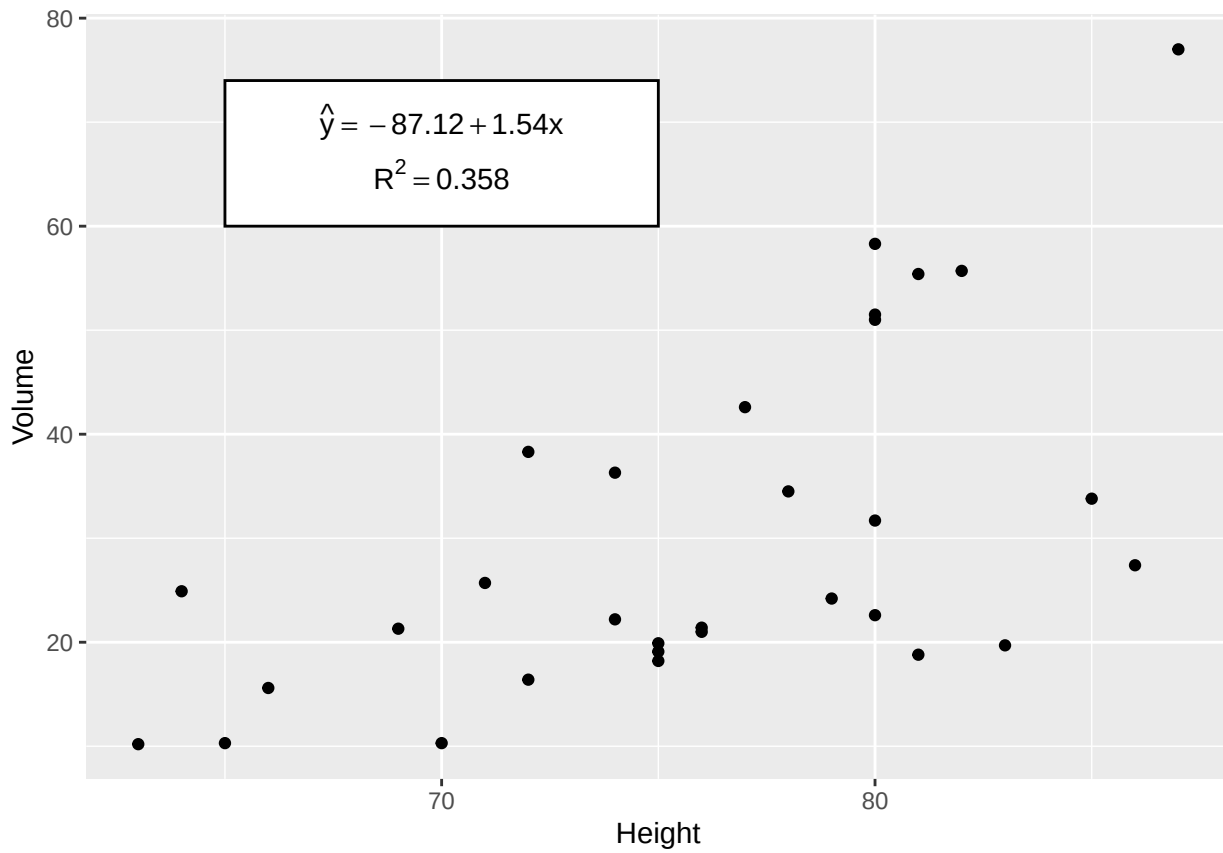
e. Add some annotation text to write the equation of the line $\hat{y}_i = -87.12 + 1.54x_i$ in the text area.

```
PlotA3 <- ggplot(trees, aes(x = Height, y = Volume)) + geom_point() +
  annotate('rect', xmin=65, xmax=75, ymin=60, ymax=74, fill='white', color='black') +
  annotate("text", x=70, y=70, label = TeX("$\\hat{y} = -87.12 + 1.54x$", output = "character"), parse =
PlotA3
```



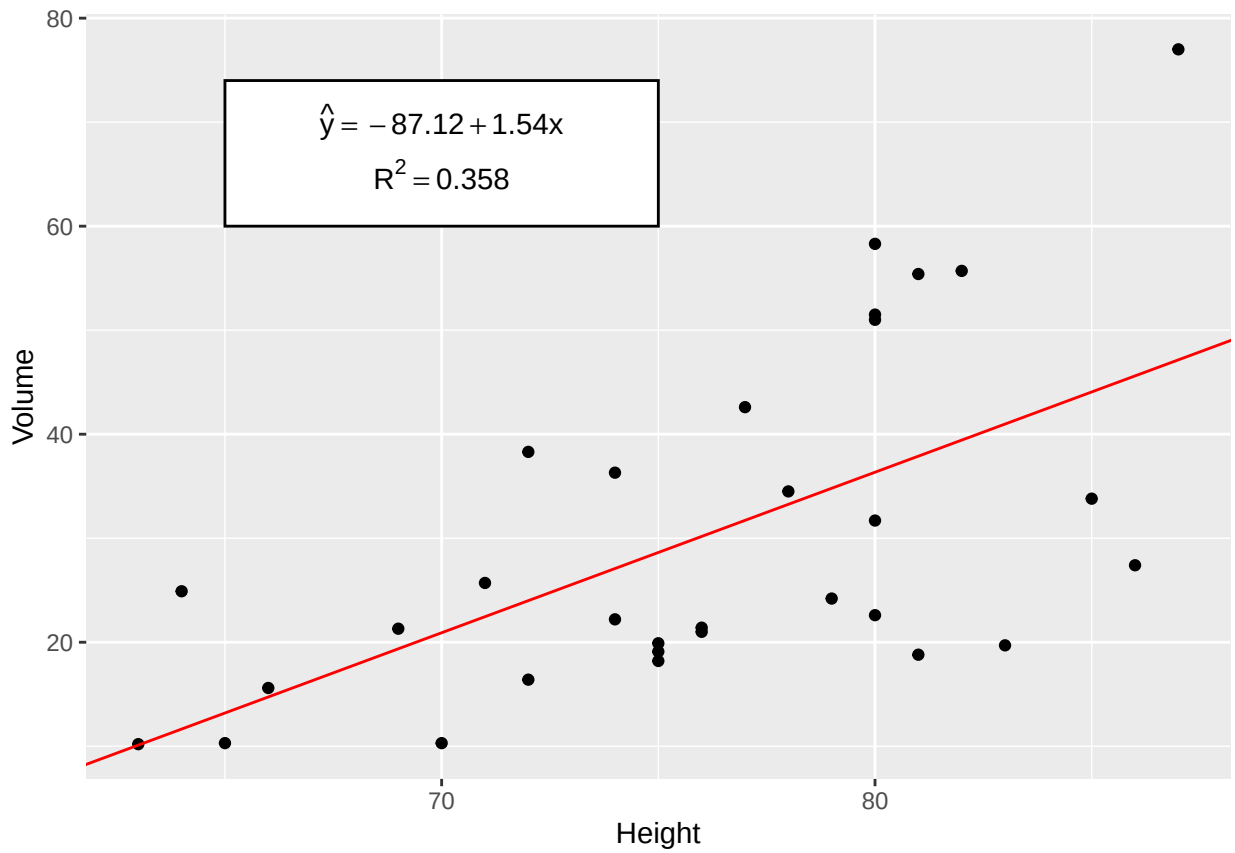
f. Add annotation to add $R^2 = 0.358$

```
PlotA4 <- ggplot(trees, aes(x = Height, y = Volume)) + geom_point() +
  annotate('rect', xmin=65, xmax=75, ymin=60, ymax=74, fill='white', color='black') +
  annotate("text", x=70, y=70, label = TeX("$\\hat{y} = -87.12 + 1.54x$", output = "character"), parse =
  annotate("text", x=70, y=65, label = TeX("$R^2 = 0.358$", output = "character"), parse = TRUE)
PlotA4
```



g. Add the regression line in red. The most convenient layer function to use is `geom_abline()`.

```
PlotA5 <- ggplot(trees, aes(x = Height, y = Volume)) + geom_point() +
  annotate('rect', xmin=65, xmax=75, ymin=60, ymax=74, fill='white', color='black') +
  annotate("text", x=70, y=70, label = TeX("$\\hat{y} = -87.12 + 1.54x$", output = "character"), parse =
  annotate("text", x=70, y=65, label = TeX("$R^2 = 0.358$", output = "character"), parse = TRUE) +
  geom_abline(intercept = regmodel$coefficients[1], slope = regmodel$coefficients[2], color = "red")
PlotA5
```



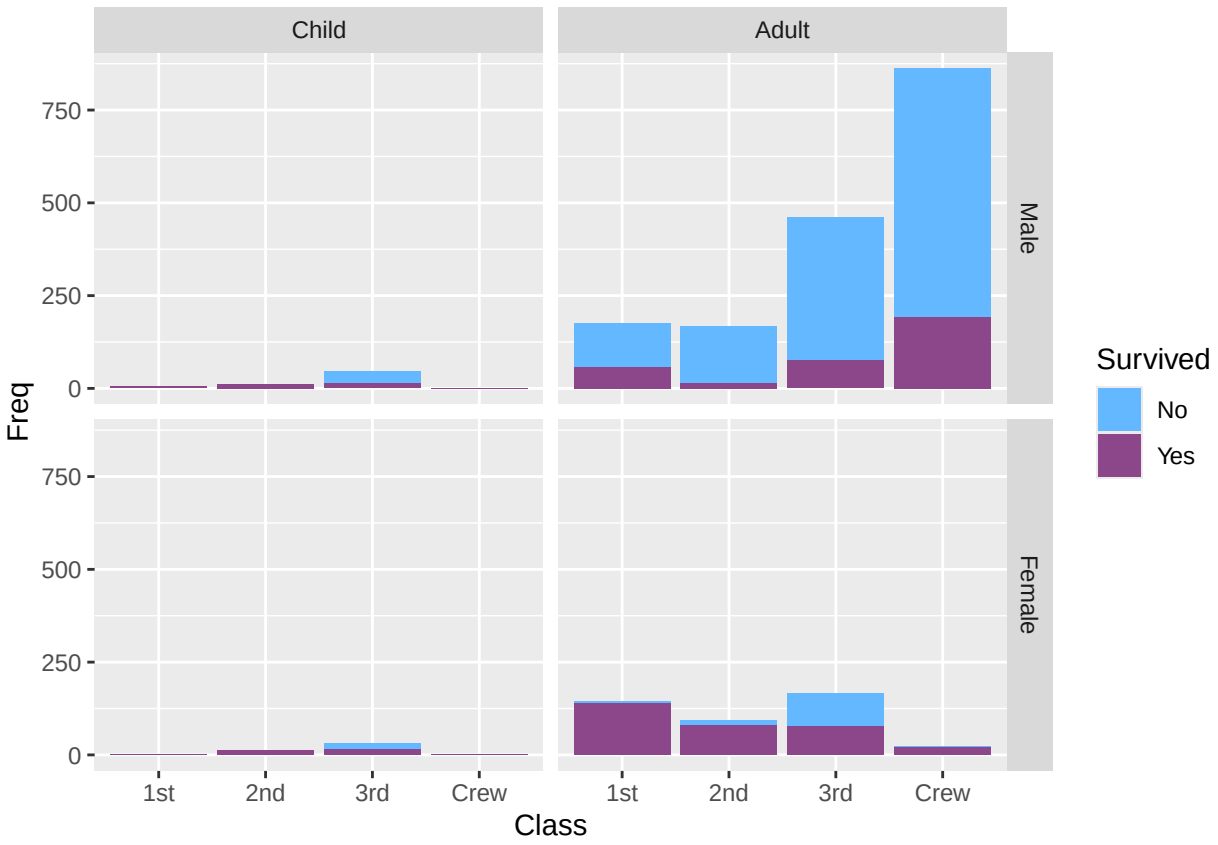
Problem 3

In `datasets::Titanic` table summarizes the survival of passengers aboard the ocean liner *Titanic*. It includes information about passenger class, sex, and age (adult or child). Create a bar graph showing the number of individuals that survived based on the passenger `Class`, `Sex`, and `Age` variable information. You'll need to use faceting and/or color to get all four variables on the same graph. Make sure that differences in survival among different classes of children are perceivable. *Unfortunately, the data is stored as a* `table`*and to expand it to a data frame, the following code can be used.*

```
‘‘‘ r
Titanic <- Titanic %>% as.data.frame()
‘‘‘
```

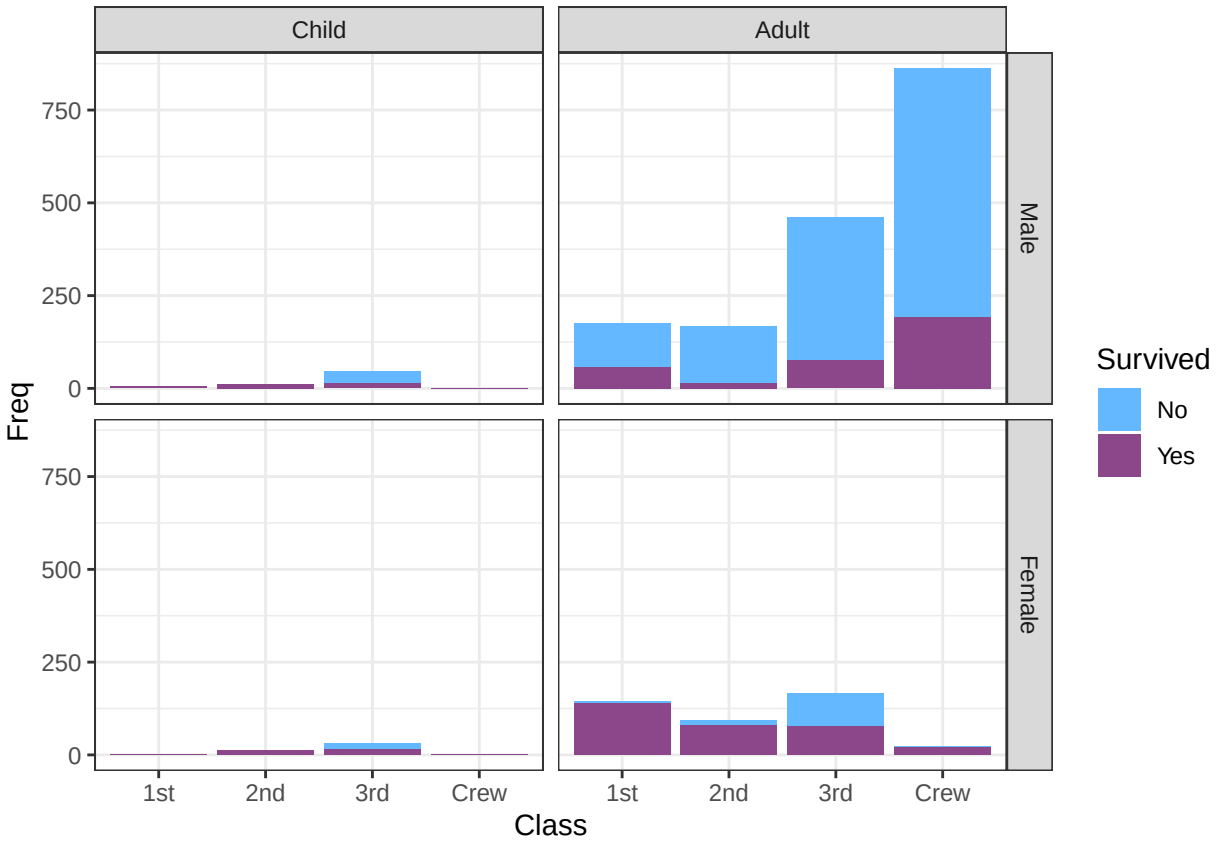
a. Make this graph using the default theme. *If you use color to denote survivorship, modify the color scheme so that a cold color denotes death.*

```r
Titanic <- Titanic %>% as.data.frame()
titanic2 <- ggplot(Titanic, aes(x = Class, y = Freq, fill = Survived)) +
  geom_bar(stat = "identity") + facet_grid(Sex ~ Age) +
  scale_fill_manual(values = c("No" = "steelblue1", "Yes" = "orchid4"))
titanic2
```



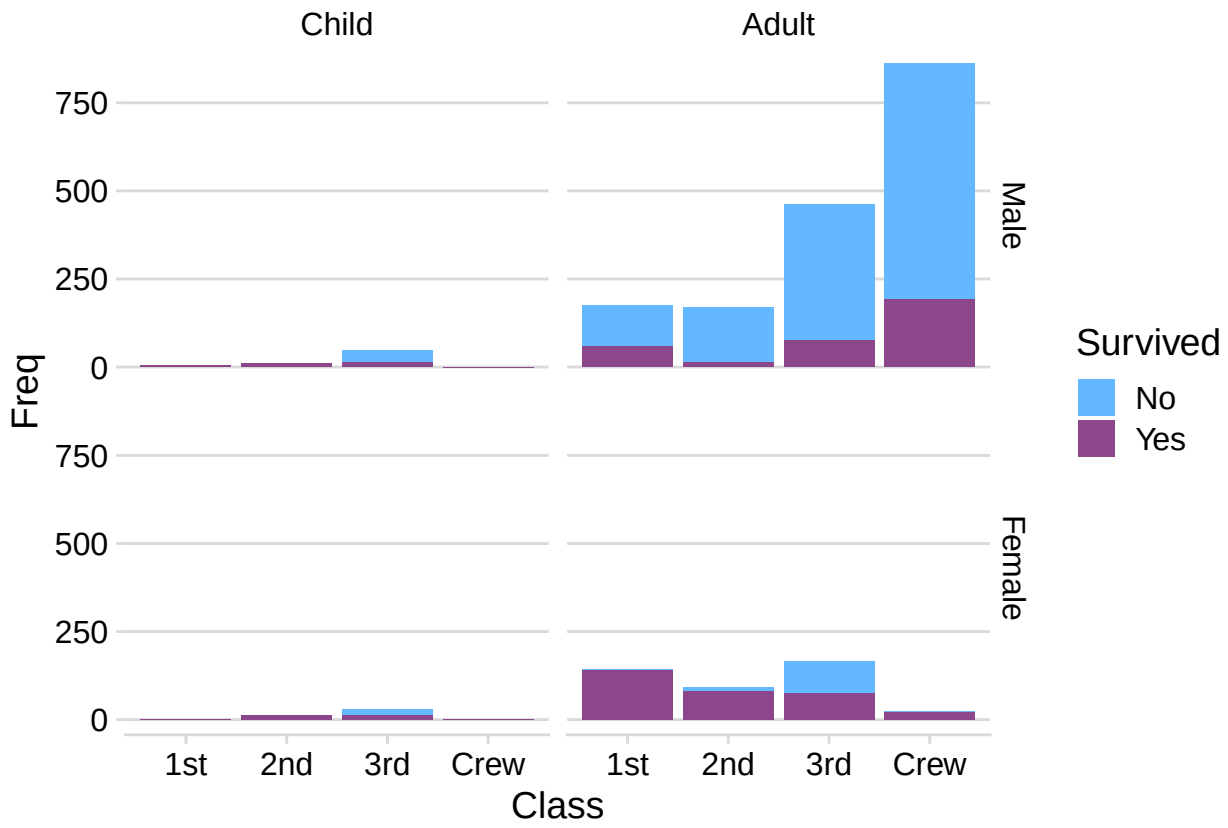
b. Make this graph using the `theme_bw()` theme.

```
(titanic3 <- titanic2 + theme_bw())
```



c. Make this graph using the `cowplot::theme_minimal_hgrid()` theme.

```
(titanic3 <- titanic2 + theme_minimal_hgrid())
```



d. Why would it be beneficial to drop the vertical grid lines? When its just the horixiontal line on the grid it makes it easieer to focus on the differences in the different categories that were comparing.